

Polycarp's Mountain Wings 🔒 DSA26 TAs

Description

Polycarp is training for the Berland Mountain Marathon. The track is described by an array of n distinct integers $a_1, a_2, \dots, a_n$ where a_i represents the altitude of the i -th point.

He wants to find the most "majestic" mountain formation to take a photo. A formation consists of three indices (i, k, j) with $1 \leq i < k < j \leq n$ that form a Mountain Wing.

- A triple (i, k, j) is a valid Mountain Wing if:
- The center k is strictly greater than both endpoints ($a_k > a_i$ AND $a_k > a_j$).
 - There must be no other index p strictly between i and j that is higher than a_k (i.e., a_k is the maximum in the range $[i, j]$).

The majesty of a mountain wing is calculated as the altitude of the peak multiplied by the sum of the altitudes of the wings:

$$\text{Score} = a_k \cdot (a_i + a_j)$$

Your task is to find the maximum Majesty Score among all valid Mountain Wings. If no such triplet exists, print 0 .

Input

The first line contains a single integer t ($1 \leq t \leq 10$), the number of test cases.

The first line of each test case contains an integer n ($3 \leq n \leq 2 \cdot 10^5$).

The second line contains n distinct integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

Output

For each test case, print a single integer representing the maximum Majesty Score.

Sample Case

Input

```
2
5
3 5 1 4 2
4
20 60 10 30
```

Copy

Output

```
35
3000
```

Copy

Comments

There are no comments at the moment.

Report an issue

Submit solution

[All submissions](#)
[Best submissions](#)

[Voting statistics](#)

[Manage tickets](#)
[Edit problem](#)
[Edit test data](#)
[Manage submissions](#)
[Clone problem](#)

✔ **Points:** 100 (partial)
⌚ **Time limit:** 1.0s
📄 **Memory limit:** 256M

✍ **Author:**
2023101016

➤ **Problem type**

✔ **Allowed languages**
C

👤 **Judges:**
[Barlon](#), [Guztavo](#),
[Leucothea](#), [Mariam](#),
[Palemar](#), [Praxidike](#),
[Tramino](#), [Zikomo](#)

 Editor

 Preview

B

I

—

H



x^2











@





1 |



Post!

Jamming Challenge

The Cyber-Defense Agency is testing a new generation of Signal Jammers. To determine which device is the best while prioritising having fun, the agency has decided to organize tournament.

Each jammer is configured to block a continuous range of integer frequencies, defined by a lower limit **L** and an upper limit **U**. The jammer effectively blocks all frequencies from **L** to **U** inclusive. The lower and upper bounds of all jammers are **distinct**.

The tournament consists of matches between every pair of jammers. The outcome of a match between Jammer **X** and Jammer **Y** is decided by **containment**:

- **Win:** Jammer **X** defeats Jammer **Y** if X's frequency range **fully contains** Y's frequency range. (i.e., **L_X** <= **L_Y** and **U_Y** <= **U_X**).
- **Draw:** If neither jammer's range fully contains the other's, the match ends in a draw.

Scoring:

- **Winner:** 2 points
- **Loser:** 0 points
- **Draw:** 1 point each

You are given the frequency limits for **N** jammers. Calculate the final score of each jammer after the tournament.

Input Format

The first line contains a single integer **T**, the number of test cases. The first line of each test case contains an integer **N**, the number of jammers. The next N lines of each test case contain two integers **L** and **U**, the frequency range for that jammer.

Output Format

For each test case, output a single line containing **N** space-separated integers, representing the total score of each jammer in the order they appear in the input.

Constraints

- 1 <= T <= 100
- 1 <= N <= 10^5
- The sum of N over all test cases does not exceed 5*10^5
- 1 <= L < U <= 10^8

Sample Input

1
3
2 3
4 9
1 10

Copy

Sample Output

1 1 4

Copy

Explanation

- **Jammer 1 [2, 3] vs Jammer 2 [4, 9]:** The ranges are disjoint (they do not overlap). Neither fully contains the other. **Draw.** (J1 gets 1 point , J2 gets 1 point).
- **Jammer 1 [2, 3] vs Jammer 3 [1, 10]:** Jammer 3's range [1, 10] fully contains Jammer 1's range [2, 3] because 1 <= 2 and 3 <= 10. **Jammer 3 Wins.** (J1 gets 0 points, J3 gets 2 points).
- **Jammer 2 [4, 9] vs Jammer 3 [1, 10]:** Jammer 3's range [1, 10] fully contains Jammer 2's range [4, 9] because 1 <= 4 and 9 <= 10. **Jammer 3 Wins.** (J2 gets 0 points, J3 gets 2 points).

Submit solution

[All submissions](#)
[Best submissions](#)

[Voting statistics](#)

[Manage tickets](#)
[Edit problem](#)
[Edit test data](#)
[Manage submissions](#)
[Clone problem](#)

✓ **Points:** 100 (partial)

⌚ **Time limit:** 2.0s

📦 **Memory limit:** 256M

 **Author:**
2023101096

➤ **Problem type**

▼ **Allowed languages**
C

 **Judges:**
[Barlon](#), [Guztavo](#),
[Leucothea](#), [Mariam](#),
[Palemar](#), [Praxidike](#),
[Tramino](#), [Zikomo](#)

- Jammer 1: 1 (Draw) + 0 (Loss) = 1
- Jammer 2: 1 (Draw) + 0 (Loss) = 1
- Jammer 3: 2 (Win) + 2 (Win) = 4

Comments

There are no comments at the moment.

Report an issue

New comment

Editor

Preview

B

I

—

H

</>

x²

“

≡

≡

@

1 |

Post!



Poshanka!

Lore

Hornet stands before the Grand Gate, the entrance to the Citadel. The **Last Judge**, a towering sentinel of white ash and gold, blocks the path. She does not merely attack; she sings a *Litany of Judgment* a rhythmic chant that manifests as waves of searing fire.

Shakra watches from the cliffside, her cartographer's eyes analyzing the Judge pattern. *"She is ancient, little princess, but her song has gaps. Her chant rises and falls in power. You cannot overpower her simply by enduring; you must find the **critical moment** the shortest possible sequence of her rhythm where the magical resonance is high enough to shatter her stance. Strike then, and strike fast. If you hesitate, the song resets."*

Scenario

From the logs of people who passed the last Judgement:

We are modeling the Judge "Litany" as a stream of integer signal values (positive and negative). To trigger a **Stagger State** in the target, we must overload their shield with a cumulative resonance of at least K .

However, the target adaptation algorithms are fast. We need to identify the **minimum time window** (shortest contiguous sequence of signals) required to reach this threshold K . If the window is too long, the target recovers before the overload occurs.

Problem Statement

You are given an array A of N integers, representing the sequence of the Judge soul-chants. You are also given an integer K , the Stagger Threshold.

Your task is to find the **length of the shortest, non-empty, contiguous subarray** of A such that the sum of the elements in that subarray is **at least** K .

If no such subarray exists, output -1 .

Note: The values in A can be negative (representing the Judge draining soul).

Input Format

- The first line contains an integer t - the number of test cases.
- For each test case:
 - The first line contains two integers N and K .
 - The second line contains N space-separated integers $A[i]$.

Output Format

For each test case, print a single integer on a new line: the length of the shortest valid subarray. If none exists, print `-1`.

Constraints

- $1 \leq t \leq 20$
- $1 \leq N \leq 2 \cdot 10^5$
- $1 \leq K \leq 10^9$
- $-10^5 \leq A[i] \leq 10^5$
- The sum of N over all test cases does not exceed $5 \cdot 10^5$.

Sample Input

```
2
6 7
2 -1 4 3 -2 5
5 15
1 2 3 4 5
```

Copy

Sample Output

Submit solution

[All submissions](#)
[Best submissions](#)

[Voting statistics](#)

[Manage tickets](#)
[Edit problem](#)
[Edit test data](#)
[Manage submissions](#)
[Clone problem](#)

✔ **Points:** 100 (partial)
⌚ **Time limit:** 1.0s
📄 **Memory limit:** 256M

✍ **Authors:**
admin, 2023102026

➤ **Problem type**

✔ **Allowed languages**
C

👤 **Judges:**
[Barlon](#), [Guztavo](#),
[Leucothea](#), [Mariam](#),
[Palemar](#), [Praxidike](#),
[Tramino](#), [Zikomo](#)

Explanation

Test Case 1: The sequence is `[2, -1, 4, 3, -2, 5]` and the threshold $K = 7$.

- Subarray `[2, -1, 4, 3]` sums to 8 (Length 4).
- Subarray `[4, 3]` sums to 7 (Length 2).
- Subarray `[4, 3, -2, 5]` sums to 10 (Length 4). The shortest valid subarray is `[4, 3]`, so the answer is 2.

Test Case 2: The sequence is `[1, 2, 3, 4, 5]` and $K = 15$. The only subarray that sums to at least 15 is the entire array itself. Length is 5.

Comments

There are no comments at the moment.

Report an issue

New comment

 Editor

 Preview

B *I* — **H** `</>` x^2 “ ” ⋮ ⋮ 🔗 🖼️ @ 📄 ?

1 |



Post!